OPEN HARDWARE TECHNICAL SPECIFICATION
Released under CERN OHL-S v2 | Prior Art Disclosure | OSF & Zenodo Deposit

Assisted Prostration Device (APD)

An Open Hardware Platform for Bi-Directional Kinetic Restoration in Repetitive Devotional and Rehabilitative Kneeling Practice

Joy Bose
Bengaluru, Karnataka, India

Date	May 2026
Licence	CERN OHL-S v2 — ohwr.org/project/cernohl
OSF DOI	https://doi.org/10.17605/OSF.IO/MQYP6
Zenodo DOI	https://doi.org/10.5281/zenodo.20351494
Repository	https://github.com/joyboseroy/apd-prostration-device
Version	1.0.0
Type	Open Hardware Specification / Prior Art Disclosure
Keywords	open hardware, prostration, Ngondro, kinetic restoration, contemplative ergonomics, CERN OHL-S v2

Abstract

Repetitive full-body prostration is a core practice in Tibetan Buddhist Ngondro (111,111 prostrations), Hindu dandavat pranama, Islamic salah, and related traditions. The cumulative biomechanical load causes patellofemoral, wrist, and lumbar injury, preventing elderly and injured practitioners from sustaining practice. No existing assistive device addresses this motion profile. The Assisted Prostration Device (APD) is a non-wearable floor-based open hardware platform providing constrained linear guided sliding across a ~1100 mm usable stroke; bi-directional kinetic energy restoration via five passive mechanical embodiments; configurable resistance geometry; optional finite-state machine cycle monitoring; and silent operation for monastery and meditation hall contexts. Typical dimensions: 2120 x 600 x 720 mm. Max safe working load: 150 kg. Released in full under CERN OHL-S v2.

Purpose of This Deposit

This document constitutes a formal public disclosure establishing a timestamped prior art record. It makes the complete APD design freely available globally, prevents any third party from patenting it, and provides a citable academic record. The full technical specification follows on subsequent pages.

Citation

Bose, J. (2026). Assisted Prostration Device (APD). Open Hardware Specification v1.0. https://doi.org/10.5281/zenodo.20351494 | https://doi.org/10.17605/OSF.IO/MQYP6

Full technical specification (13 sections) follows on subsequent pages.

Assisted Prostration Device (APD)

An Open Hardware Platform for Bi-Directional Kinetic Restoration in Repetitive Devotional and Rehabilitative Kneeling Practice

Joy Bose
Bengaluru, Karnataka, India
Licence: CERN OHL-S v2 | Repository: https://github.com/joyboseroy/apd-prostration-device

Author	Joy Bose
Address	Bengaluru, Karnataka, India
Date of Deposit	May 2026
Licence	CERN Open Hardware Licence v2 — Strongly Reciprocal (CERN OHL-S v2)
Repository	https://github.com/joyboseroy/apd-prostration-device
OSF DOI	https://doi.org/10.17605/OSF.IO/MQYP6
Zenodo DOI	To be assigned upon Zenodo deposit
Version	1.0.0
Document Type	Open Hardware Specification / Prior Art Disclosure
Keywords	open hardware, assistive device, prostration, Ngondro, contemplative ergonomics, kinetic restoration, CERN OHL-S v2

Abstract

Repetitive full-body prostration is a core practice in Tibetan Buddhist Ngondro (111,111 prostrations), Hindu dandavat pranama, Islamic salah, and related traditions. The cumulative biomechanical load causes patellofemoral, wrist, and lumbar injury, preventing elderly and injured practitioners from sustaining practice. No existing assistive device addresses this motion profile. The Assisted Prostration Device (APD) is a non-wearable floor-based open hardware platform providing: constrained linear guided sliding across a ~1100 mm usable stroke; bi-directional kinetic energy restoration via five passive mechanical embodiments (elastomeric spring, gas strut, counterweight-pulley, cam-based nonlinear resistance, and magnetic eddy-current damping); configurable resistance geometry; optional finite-state machine cycle monitoring; and silent operation for monastery and meditation hall contexts. Typical dimensions: 2120 x 600 x 720 mm. Max safe working load: 150 kg. Released under CERN OHL-S v2.

Purpose of This Deposit

This document constitutes a formal public disclosure of the APD open hardware design, establishing a timestamped prior art record. It makes the complete design freely available to practitioner communities globally, prevents any third party from patenting it, and provides a citable academic record. The full technical specification follows on subsequent pages.

Citation

Bose, J. (2026). Assisted Prostration Device (APD): An Open Hardware Platform for Bi-Directional Kinetic Restoration in Repetitive Devotional and Rehabilitative Kneeling Practice. Open Hardware Specification v1.0. https://github.com/joyboseroy/apd-prostration-device

---

1. TITLE OF THE INVENTION

Assisted Prostration Device with Bi-Directional Kinetic Restoration Sliding Platform for Repetitive Devotional and Rehabilitative Kneeling Motions

2. FIELD OF THE INVENTION

The present invention relates to assistive mechanical devices, biomechanical ergonomics, and contemplative ergonomics. More specifically, it relates to a non-wearable, floor-based, modular platform system providing a guided, low-friction linear forward-sliding motion path combined with a bi-directional mechanical energy-storage and return assembly — herein termed the Kinetic Restoration Subsystem (KRS). The device is ergonomically optimised for prolonged high-repetition devotional practices including but not limited to Buddhist Ngondro prostrations, Hindu dandavat pranama, Islamic salah rakat cycles, and equivalent clinical rehabilitative kneeling therapies.

3. BACKGROUND OF THE INVENTION

Repetitive full-body prostration constitutes a core physical and spiritual practice across numerous global contemplative and devotional traditions. In Tibetan Buddhist Ngondro preliminary practice, a practitioner traditionally performs 111,111 full-body prostrations as a foundational accumulation. Each repetition involves transitioning from standing to kneeling, lowering the entire torso prone with arms fully extended forward, returning to kneeling, and ascending to standing. The aggregate biomechanical load across thousands of such cycles creates substantial musculoskeletal risk:

- Repeated eccentric loading of the knee joint during descent causes progressive patellofemoral stress, cartilage wear, and meniscal degradation.
- Wrist, palm, and glenohumeral joints endure repetitive impact forces during floor contact, causing tendinitis and carpal tunnel symptoms.
- Severe lower back and hip flexor fatigue limits session duration, particularly for elderly, injured, or frail practitioners.
- Practitioners with pre-existing joint conditions are often unable to sustain complete doctrinal practice.

Existing prior art categories do not address this specific biomechanical and ritual need:

- Industrial and military exoskeletons (US10,369,071B2; EP3228284A1): Wearable, power-dependent, designed for bipedal gait and vertical load-bearing. Incompatible with floor-level prone-extension geometry.
- Clinical rehabilitation aids (US6,393,641B1): Designed for one-time sit-to-stand transitions; lack cyclic energy storage for continuous high-repetition use.
- Traditional prostration boards: Wooden boards with hand-held sliders reduce surface friction but provide zero joint offloading and no ascent assistance.
- Electronic counters and phase-recognition systems (CN106805385A): Provide posture detection or non-assistive counting without physical load mitigation.

No prior art discloses a non-wearable floor-based platform combining constrained prone-extension devotional geometry, bi-directional kinetic energy storage, configurable resistance geometry, safety features, and optional integrated cycle monitoring in a system optimised for the continuous full-cycle biomechanical demands of kneeling-to-prone-to-kneeling devotional sequences. This gap defines the present invention.

4. OBJECTS OF THE INVENTION

1. To significantly reduce cumulative musculoskeletal loading on the patellofemoral joints, lumbar spine, wrists, and shoulder girdle during high-repetition prostration cycles.
2. To provide a completely non-wearable, self-contained floor unit requiring no body harness, personalised fitting, or user-specific calibration.
3. To implement a passive or semi-passive mechanical energy-capture mechanism that harvests kinetic energy during forward body extension and redistributes it to assist the rearward ascent phase without requiring external electrical power.
4. To provide configurable resistance geometry through adjustable anchor positions, interchangeable spring or strut modules, and cam-based nonlinear resistance elements enabling adaptation for practitioners of varying body weight and physical condition.
5. To fully preserve the doctrinal motion sequence of the prostration cycle without abbreviation, substitution, or mechanical distortion.
6. To embed an optional unobtrusive digital counting system employing finite-state machine logic to eliminate false counts and support long-form practice accumulation.
7. To provide a modular framework capable of rapid folding and toolless disassembly for storage and transport across domestic and monastic spaces.
8. To incorporate safety features including anti-tip stabilisers, soft-lock travel limits, damped return speed, quick-release disengagement, and knee alignment guides ensuring safe operation across a range of practitioner abilities.

5. SUMMARY OF THE INVENTION

The present invention provides an Assisted Prostration Device (APD) comprising five principal subsystems: Subsystem A (Base Frame Assembly), Subsystem B (Sliding Platform Assembly), Subsystem C (Bi-Directional Kinetic Restoration Subsystem), Subsystem D (Support Rail Assembly), and Subsystem E (Electronic Monitoring Module, optional). The device operates without mandatory external electrical power. The practitioner kneels on the fixed base pad (12), rests hands or forearms on the sliding platform (13), and performs forward extension. The platform (13) travels forward along constrained longitudinal channel rails (17) distributing upper-body load and reducing wrist impact. The Kinetic Restoration Subsystem (24) stores energy during forward travel and releases it during the ascent phase, providing partial mechanical lift. The support rails (11) provide grip stability throughout. The complete doctrinal motion sequence is preserved while eccentric joint loading is substantially reduced across high-repetition practice.

6. DETAILED DESCRIPTION OF THE INVENTION

6.1 System Architecture and Typical Dimensions

The APD is organised into five operational subsystems as described above. Reference numerals used throughout correspond to those in the accompanying Figures 1 through 10. Typical overall dimensions as derived from engineering drawings are: Overall Length 2120 mm; Overall Width 600 mm; Overall Height to top of rail 720 mm; Sliding Stroke (usable) approximately 1100 mm; Device Weight 28 to 35 kg depending on configuration; Maximum Safe Working Load 150 kg.

All dimensions are typical and may vary based on user requirements and manufacturing tolerances of plus or minus 5 mm. Materials are stated as typical specifications; equivalent substitute materials providing equivalent mechanical properties are within the scope of the invention.

6.2 Subsystem A — Base Frame Assembly (Figure 1, Figure 9)

The base frame chassis (20) is a rigid rectangular low-profile frame fabricated from powder-coated steel in a preferred embodiment, with typical overall dimensions of 2120 mm (length) x 600 mm (width) x approximately 85 to 95 mm (height from floor to platform surface, adjustable). The base frame chassis (20) is fitted with four adjustable levelling feet (23) of steel and rubber construction providing 10 to 25 mm of height adjustment and high-friction floor contact.

The rearward zone of the base frame houses the fixed kneeling pad (12), a cushioned surface of approximate dimensions 460 mm (length) x 600 mm (width) x 40 mm (thickness), covered with high-density EVA foam with polyurethane cover providing patellofemoral impact isolation. The forward zone houses the longitudinal channel rails (17) of anodized aluminium alloy construction within which the sliding platform (13) travels. The rear travel stop (21) and front rubber bumper (22) of natural rubber construction bound the sliding stroke at each end. A side access panel (25) on both sides of the base frame provides maintenance access to the Kinetic Restoration Subsystem (24) components housed within. Anti-tip stabiliser geometry is incorporated into the base frame lateral profile.

6.3 Subsystem B — Sliding Platform Assembly (Figure 6)

The sliding platform assembly (13) is the primary user-contact structure. It is a rigid planar deck of anodized aluminium alloy construction, rolling on six low-friction rolling elements (16) of sealed ball-bearing wheels with polyurethane tread, approximately 40 mm in diameter with 2RS sealed bearings, each rated to approximately 80 kg dynamic load. The rolling elements (16) run within the longitudinal channel rails (17), constraining motion exclusively to the linear longitudinal axis and eliminating lateral skewing.

The upper surface of the sliding platform carries three distinct contact zones: forearm support pads (14), a pair of closed-cell EVA foam pads of approximately 250 mm x 120 mm x 40 mm each positioned for forearm loading during the extension phase; hand contact surface (15), a high-density EVA foam pad with anti-slip, skin-friendly surface of approximately 260 mm x 600 mm x 40 mm positioned at the forward end for hand and palm contact; and the main deck surface (13) of glass-filled nylon or ABS with ribbed anti-slip texture providing a stable low-friction travel environment for the forearms during extension. All contact surface pads are replaceable modules secured with concealed fasteners for maintenance and hygiene.

The quick-release locking knob (18) of glass-filled nylon allows the sliding platform position to be manually locked at any point along the stroke for maintenance or storage. The front rubber bumper (22) absorbs end-of-travel shock. The rear travel stop (21) limits rearward travel, maintaining clearance from the fixed kneeling pad (12).

6.4 Subsystem C — Bi-Directional Kinetic Restoration Subsystem (Figure 5)

The Kinetic Restoration Subsystem (24) is the primary mechanical engine of the invention. It is housed within the base frame chassis (20) and accessed via the side access panels (25). It stores potential energy as the sliding platform (13) travels forward under body weight during the forward extension phase (Stage 2 of the four-stage operation sequence) and returns that energy to provide rearward assist during the return phase (Stage 4). The device operates passively without electrical power in all KRS embodiments described below.

Operation Sequence (Figure 8): Stage 1 — Kneeling start position, springs and struts (261, 264) in neutral, platform at rear stop (21), no stored energy. Stage 2 — Forward extension, body weight drives platform (13) forward, springs (261) stretch and strut (264) compresses, kinetic energy captured in KRS (24). Stage 3 — Transition at full extension, direction reverses, stored energy begins to assist return. Stage 4 — Return, KRS (24) discharges stored energy pulling platform (13) rearward, user returns to kneeling position with reduced effort, platform reaches rear stop (21), one complete cycle.

Embodiment C1 — Elastomeric / Extension Spring (261):

One or more high-carbon steel extension springs or industrial-grade elastomeric cords (261) are routed through a central protective channel within the base frame (20). Each spring (261) connects a fixed anchor bracket (263) on the underside of the sliding platform (13) to an adjustable anchor pin (262) on the base frame chassis (20). As the platform (13) travels forward, the distance between bracket (263) and anchor pin (262) increases, storing potential energy. The longitudinal position of anchor pin (262) is adjustable across pre-drilled holes in the chassis (20) to tune pre-tension and force profile. Return force is calibrated to 25 to 40 percent of the practitioner's estimated upper-body mass.

Embodiment C2 — Gas Compression Strut Cylinder (264):

Dual gas compression strut cylinders (264) of painted steel construction are pivotally pinned at an oblique angle relative to the horizontal travel plane within the base frame (20).

During forward stroke, the piston rod (265) retracts, compressing the internal nitrogen volume to build progressive resistance. The oblong mounting pivot matrix (266) of aluminium alloy allows the pivot insertion angle to be altered, adapting mechanical advantage to practitioner mass. Struts are rated between 200 N and 400 N force; interchangeable struts of graduated ratings accommodate different user weight profiles. Damped return speed limits maximum rearward retraction velocity preventing unsafe snap-back.

Embodiment C3 — Counterweight and Pulley:

A continuous high-tensile cable or Dyneema line routes from the sliding platform (13) over a forward pulley to a counterweight suspended within an enclosed safety housing at the nose of the device. This delivers a position-independent return force profile across the full stroke length.

Embodiment C4 — Cam-Based Nonlinear Resistance (Figure 5, Eccentric Cam Plate 267):

The eccentric cam plate (267) of aluminium alloy is keyed to the rolling element axle (22). A cantilever arm (268) tracks the changing profile radius of the cam (267) during forward extension. The variable radius alters the instantaneous mechanical leverage of the secondary leverage spring component (269), matching the force curve to the practitioner's biomechanical extension profile. Moving the anchor pin (262) to different holes on the cam plate (267) changes the effective leverage ratio: Hole A providing higher assistance and lower resistance during the forward stroke, Hole B providing balanced medium assistance, and Hole C providing lower assistance and higher resistance. This nonlinear profile provides higher resistance in mid-stroke and reduced resistance at stroke initiation and completion.

Embodiment C5 — Magnetic Eddy-Current Damping:

Permanent magnet arrays mounted on the sliding platform (13) interact with aluminium conductor plates fixed to the base frame (20), generating velocity-proportional braking force during forward extension and reducing shock loading at floor contact. This embodiment provides silent, maintenance-free resistance with no moving mechanical parts subject to wear.

6.5 Subsystem D — Support Rail Assembly (Figure 7)

Two vertical support rails (11) of aluminium alloy 6061-T6 tube construction, powder-coated finish, are anchored to vertical support rail base brackets (30) of mild steel with powder-coat finish bolted to the rearward end of the base frame (20). The rails (11) carry rail grips (19) of thermoplastic rubber (TPE) providing non-slip hand contact surfaces. The two rails (11) are joined at their upper ends by a crossbar (31) of aluminium alloy, forming an inverted U-shaped safety cage with overall width of approximately 600 mm and typical grip height of 720 mm.

Rail height is adjustable via a pin-locking adjustment matrix (32) of stainless steel insert construction, providing height settings between approximately 520 mm (low position for elderly and rehabilitation users) and 720 mm (standard and tall user position) in discrete increments. Quick-release locking knobs (18) of glass-filled nylon allow rapid rail removal or folding for transport. The crossbar (31) adds torsional rigidity and provides an optional alignment reference during the kneeling start position. All rail edges are radiused and surfaces are powder-coated or anodized for silent, durable operation.

6.6 Subsystem E — Electronic Monitoring Module (Figure 10, component 40)

An optional electronic monitoring module (40) integrates into the APD. The hardware stack comprises a force-sensitive resistor array (42) embedded beneath the kneeling pad (12), connected to a microcontroller unit (44), driving a digital display (46) mounted on the support rail (11) at eye level, with a Bluetooth Low Energy transceiver (48) for companion application connectivity. The module is powered by an internal rechargeable lithium-polymer cell and features a hardware reset toggle for zero-setting or countdown-mode configuration.

The counting algorithm employs a finite-state machine (FSM) with two-phase threshold logic (Figure 12 / Section 11). State 0 represents the baseline condition with pad pressure above threshold P_high. Transition to State 1 occurs when pressure drops below P_low, validating forward extension. Transition to State 2 occurs when pressure recovers above P_high, validating return to kneeling. The count increments only on a successful consecutive 0-to-1-to-2 state sequence. Non-sequential events such as knee repositioning, robe adjustment, or partial extension fail the state criteria and are discarded as noise.

6.7 Modes of Operation

Mode 1 — Passive Assistive Slide:

Completely unpowered. The KRS (24) autonomously manages shock absorption and partial return assistance. Suitable for all practitioners; primary mode of use.

Mode 2 — Guided Rail Support:

Practitioner maintains continuous hand contact with support rails (11) throughout the complete motion range. Offloads significant lower-extremity load. Designed for elderly practitioners, acute joint damage, or rehabilitative clinical contexts.

Mode 3 — Monitored Accumulation:

Electronic monitoring module (40) active. Cycle counts displayed on display (46). Countdown alerts on session target completion. No gamified feedback; steady numerals only, consistent with contemplative focus.

6.8 Contemplative Design Principles

The APD incorporates several design principles specific to contemplative practice contexts that distinguish it from industrial or rehabilitation device paradigms. Silent operation is achieved through precision 2RS sealed bearings in rolling elements (16), polymer track liners in rails (17), rubber vibration-damping in bumpers (22) and levelling feet (23), maintaining near-zero mechanical noise suitable for monastery and meditation hall environments. Visual minimalism is maintained through matte powder-coated and anodized finishes in dark tones. The device preserves the complete four-stage doctrinal motion sequence of prostration cycles without abbreviation, substitution, or distortion of the gesture, maintaining the contemplative and ritual integrity of the practice.

7. LIST OF REFERENCE NUMERALS

(Reconciled with Figures 1 through 10)

11	Vertical Support Rail (Left and Right) — Aluminium Alloy 6061-T6, Powder Coated
12	Fixed Kneeling Pad — EVA Foam / PU Cover, 460 x 600 x 40 mm typical
13	Sliding Platform Assembly (Main Deck) — Anodized Aluminium Alloy
14	Forearm Support Pads (Pair) — Closed-Cell EVA Foam, 250 x 120 x 40 mm each
15	Hand Contact Surface — High-Density EVA Foam / PU Cover, Anti-Slip, 260 x 600 x 40 mm
16	Low-Friction Rolling Elements (Inside Channel) — PU Tread on 2RS Sealed Ball Bearings, ~40 mm dia.
17	Longitudinal Channel Rail — Anodized Aluminium Alloy (mounted in Base Frame)
18	Quick-Release Locking Knob — Glass-Filled Nylon
19	Rail Grip (Polymer) — Thermoplastic Rubber (TPE)
20	Base Frame Chassis — Powder Coated Steel
21	Rear Travel Stop
22	Front Rubber Bumper — Natural Rubber
23	Adjustable Levelling Feet (4 Nos.) — Steel and Rubber, 10–25 mm adjustment range
24	Kinetic Restoration Subsystem — Inside Base Frame (Subsystem C)
25	Side Access Panel (Both Sides)
30	Vertical Support Rail Base Bracket — Mild Steel, Powder Coated
31	Crossbar (Rail Top Connector) — Aluminium Alloy
32	Pin-Locking Adjustment Matrix — Stainless Steel Insert
40	Electronic Monitoring Module (Optional — Subsystem E)
42	Force-Sensitive Resistor (FSR) Array — embedded in kneeling pad (12)
44	Microcontroller Unit (MCU) — e.g. ATmega328P or equivalent
46	Digital Display (LED / e-Paper) — mounted on rail (11) at eye level
48	Bluetooth Low Energy (BLE) Transceiver
261	Extension Spring / Elastomeric Cord — High Carbon Steel / Industrial Elastomer
262	Adjustable Anchor Pin — in Base Frame Chassis (20)
263	Fixed Anchor Bracket — on underside of Sliding Platform (13)
264	Gas Compression Strut Cylinder — Steel, Painted, 200–400 N rated
265	Piston Rod Assembly — within Gas Strut (264)
266	Oblong Mounting Pivot Matrix — Aluminium Alloy
267	Eccentric Cam Plate — Aluminium Alloy, with Holes A / B / C for leverage tuning
268	Cantilever Arm — Steel

269	Secondary Leverage Spring Component — High Carbon Steel

8. TYPICAL DIMENSIONS AND MATERIAL SPECIFICATIONS

(From engineering drawings, Figures 9 and 10. All dimensions in millimetres. Tolerances plus or minus 5 mm unless stated otherwise.)

Overall Length	**2120 mm**
Overall Width	**600 mm**
Overall Height (to top of rail)	**720 mm**
Platform Height from Floor (adjustable)	**85 to 95 mm**
Sliding Stroke (usable)	**~1100 mm**
Kneeling Pad Size (L x W x T)	**460 x 600 x 40 mm**
Forearm Pad Size each (L x W x T)	**250 x 120 x 40 mm**
Hand Pad Size (L x W x T)	**260 x 600 x 40 mm**
Rail Height — Low Position (grip)	**~520 mm**
Rail Height — High Position (grip)	**~720 mm**
Ground Clearance (adjustable range)	**10 to 25 mm**
Device Weight	**28 to 35 kg (by configuration)**
Maximum Safe Working Load	**150 kg**
Rolling Element Diameter	**~40 mm**
Rolling Element Bearing Type	**2RS Sealed Ball Bearing**
Rolling Element Load Capacity (each)	**~80 kg dynamic**

9. INDICATIVE CLAIMS

(Note: The following indicative claims are included as a record of the inventive scope. This document is an open hardware specification, not a patent application. The design is released under CERN OHL-S v2.)

INVENTIVE SCOPE (Open Hardware — Not a Patent Application):

1. An assisted prostration and rehabilitative kneeling device comprising: a rigid ground-engaging base frame chassis (20) having a dedicated fixed kneeling pad (12) at its rearward terminus; a sliding platform assembly (13) translationally coupled within longitudinal channel rails (17) fixed to said base frame (20), wherein the sliding platform (13) is constrained exclusively to linear travel along the longitudinal axis of said base frame; and a bi-directional kinetic restoration subsystem (24) mechanically interconnecting said sliding platform (13) to said base frame (20), configured to capture and store energy during forward linear extension

of said sliding platform under body weight and to discharge said stored energy to apply a rearward assistance force during the retraction phase of a prostration cycle.

2. The device of claim 1, wherein said bi-directional kinetic restoration subsystem (24) comprises a system selected from the group consisting of: high-carbon steel extension springs or elastomeric cords (261) with adjustable anchor pin (262) positions in the base frame (20); gas compression strut cylinders (264) with oblique mounting pivot matrix (266); counterweight-and-pulley cable systems providing position-independent return force; cam-based nonlinear resistance mechanisms comprising an eccentric cam plate (267), cantilever arm (268), and leverage spring (269); and magnetic eddy-current damping arrays providing velocity-proportional braking.
3. The device of claim 1, wherein the return force of the bi-directional kinetic restoration subsystem (24) is calibrated to offset between 25 percent and 40 percent of an adult practitioner's upper-body mass during the ascent phase.
4. The device of claim 1, further comprising a support rail assembly comprising a pair of vertical support rails (11) of aluminium alloy construction anchored via base brackets (30) to the rearward end of said base frame (20), joined at their upper ends by a crossbar (31) forming an inverted structural cage, wherein rail height is adjustable via a pin-locking adjustment matrix (32) between approximately 520 mm and 720 mm grip height, and wherein quick-release locking knobs (18) permit rapid rail removal for transport.
5. The device of claim 1, further comprising configurable resistance geometry comprising one or more of: adjustable anchor pin (262) positions along the longitudinal axis of the base frame (20); interchangeable spring or strut modules of graduated force ratings; and an eccentric cam plate (267) with selectable pivot hole positions (A, B, C) on a cantilever arm (268) modifying the leverage ratio and resistance curve as a function of sliding platform position.
6. The device of claim 1, further comprising safety elements comprising one or more of: anti-tip stabiliser geometry incorporated into the base frame (20) lateral profile; a rear travel stop (21) and front rubber bumper (22) bounding the sliding stroke; adjustable levelling feet (23) providing floor stability; damped return speed via gas strut (264) damping limiting maximum rearward retraction velocity; and quick-release locking knob (18) permitting manual position lock.
7. The device of claim 1, further comprising an electronic monitoring module (40) comprising a force-sensitive resistor array (42) embedded beneath the fixed kneeling pad (12), a microcontroller unit (44), and a digital display (46) mounted on the support rail (11), configured to detect and record complete prostration cycles employing a finite-state machine that increments a cycle count only upon a consecutive State 0 to State 1 to State 2 transition sequence, wherein State 0 represents pad pressure above a high threshold P_{high}, State 1 represents pad pressure below a low threshold P_{low} corresponding to forward extension, and State 2 represents pad pressure recovery above P_{high} corresponding to return to kneeling.
8. The device of claim 7, wherein the electronic monitoring module (40) further comprises a Bluetooth Low Energy transceiver (48) for session data transmission to a companion application, a rechargeable lithium-polymer power cell, and a hardware reset toggle for countdown-mode configuration.
9. The device of claim 1, wherein the sliding platform assembly (13) carries at its upper surface: a pair of forearm support pads (14) of closed-cell EVA foam; a hand contact surface (15) of high-density EVA foam with anti-slip finish; and a main deck surface of glass-filled nylon with ribbed anti-slip texture; wherein all contact surface pads are replaceable modules secured with concealed fasteners.
10. The device of claim 1, wherein the base frame chassis (20) is fabricated from powder-coated steel with four adjustable levelling feet (23) providing 10 to 25 mm height adjustment, a side access panel (25) on both sides providing maintenance

access to the kinetic restoration subsystem (24), and wherein overall device dimensions are approximately 2120 mm in length, 600 mm in width, and 720 mm in height to the top of the support rails (11).

11. A method of providing guided mechanical assistance during repetitive full-body devotional or therapeutic prostration sequences using the device of claim 1, the method comprising four stages: Stage 1, positioning a practitioner in a kneeling stance upon the fixed kneeling pad (12) with hands or forearms resting on the sliding platform (13) at the rear travel stop (21) with the kinetic restoration subsystem (24) in neutral; Stage 2, executing a forward body extension driving the sliding platform (13) forward along the channel rails (17), thereby loading the kinetic restoration subsystem (24) while transferring upper body weight away from wrists and spine; Stage 3, at full prone extension pausing briefly in the doctrinal position, then initiating body-ascent contraction causing the kinetic restoration subsystem (24) to begin discharging stored energy; and Stage 4, returning to the kneeling stance with partial mechanical assistance from the kinetic restoration subsystem (24) as the sliding platform (13) is drawn rearward to the rear travel stop (21), completing one full cycle.

10. ABSTRACT

An assisted prostration device (APD) for facilitating high-repetition kneeling and full-body devotional or rehabilitative prostration comprises a non-wearable floor-based platform system. The device includes a rigid powder-coated steel base frame chassis (20) with a fixed cushioned kneeling pad (12), a sliding platform assembly (13) mounted on low-friction sealed ball-bearing rolling elements (16) constrained within anodized aluminium longitudinal channel rails (17), a bi-directional kinetic restoration subsystem (24) storing energy during forward body extension and releasing it during the ascent phase, and a pair of adjustable aluminium alloy support rails (11) joined by crossbar (31). During forward extension, the sliding platform (13) travels linearly absorbing wrist and arm impact while the kinetic restoration subsystem (24) loads across an approximately 1100 mm usable sliding stroke. During the ascent phase, the subsystem (24) discharges stored energy providing partial return assistance and reducing eccentric patellofemoral and lumbar loading. Configurable resistance geometry via adjustable anchor positions (262), interchangeable spring or strut modules (261, 264), or eccentric cam plate (267) with selectable leverage positions (A, B, C) allows adaptation for different users. Safety features including anti-tip geometry, travel stops (21, 22), adjustable levelling feet (23), and damped return speed ensure safe operation across a maximum safe working load of 150 kg. An optional electronic monitoring module (40) uses a force-sensitive resistor array (42) and finite-state machine threshold logic to record valid prostration cycles and transmit session data via Bluetooth transceiver (48). The device operates without external electrical power in its passive configuration, incorporates near-silent 2RS sealed bearing mechanics and powder-coated surfaces for monastery and meditation hall use, and preserves the complete four-stage doctrinal motion sequence of the prostration cycle. Applications include Buddhist Ngondro preliminary practice, Hindu and Islamic devotional prostration, elderly mobility support, occupational therapy, and rehabilitative kneeling therapy. Typical overall dimensions: 2120 mm (L) x 600 mm (W) x 720 mm (H), device weight 28 to 35 kg.

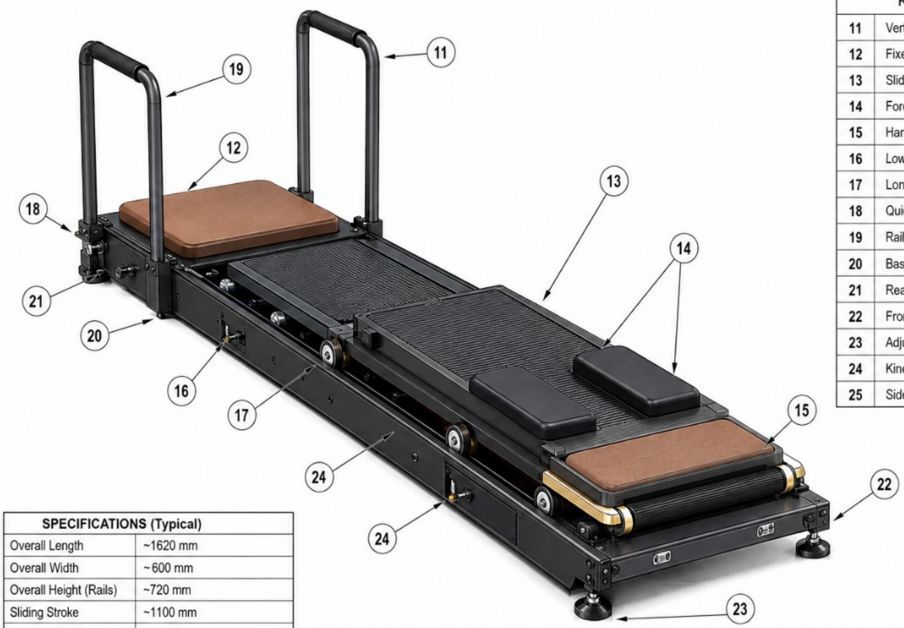
11. BRIEF DESCRIPTION OF DRAWINGS

Figures 1 through 10 are engineering drawings prepared by the inventor as reference illustrations. Formal line drawings conforming to IPO requirements to be filed with the Complete Specification. Reference numerals correspond to Section 7.

- Figure 1: Perspective view of the APD in deployed configuration showing all subsystems and specifications table.
- Figure 2: Side perspective view (left side) showing primary components, linear motion path, sliding stroke of approximately 1100 mm, and reference numerals.
- Figure 3: Top plan view (labelled) showing all components from above, sliding direction, and dimension and specification tables.
- Figure 4: Exploded perspective view showing subsystem breakdown with all components separated and labelled, including Kinetic Restoration Subsystem (C) components visible inside base frame.
- Figure 5: Inside view of Kinetic Restoration Subsystem (C) showing all embodiment components (261, 264, 267, 268, 269, 262, 266) with operation sequence diagrams for forward stroke (energy capture) and return stroke (energy release), and eccentric cam plate (267) resistance tuning detail with Hole A, B, C positions.
- Figure 6: Sliding Platform Assembly (Subsystem B) construction details showing rolling elements (16), channel rail (17), contact surface zones (14, 15), cross-section view, contact surface materials table, rolling element specifications, and quick-release and travel feature details.
- Figure 7: Support Rail Assembly (Subsystem D) showing rail height adjustment detail in low and high positions (520 mm and 720 mm), front view with overall dimensions, features and functions summary, and material specifications table.
- Figure 8: Operation sequence of APD showing full prostration cycle in four stages (Stage 1 Kneeling Start, Stage 2 Forward Extension Energy Capture, Stage 3 Transition Direction Change, Stage 4 Return Energy Release) with system state annotations and kinetic energy flow legend.
- Figure 9: Overall dimensions and orthographic views (Top, Side, Front Rear End, Front Forward End, Side with stroke indication, Isometric overview) with complete dimension summary table and materials table.
- Figure 10: Exploded view and component breakdown showing main exploded view with all reference numerals, sub-assembly groups for all five subsystems, fasteners and small parts summary, and material summary table.
- Figure 11 (to be prepared for Complete Specification): Finite-State Machine state diagram for Electronic Monitoring Module (40) as detailed in Section 12.

FIG. 1 – PERSPECTIVE VIEW OF ASSISTED PROSTRATION DEVICE (APD)

Overall Architecture



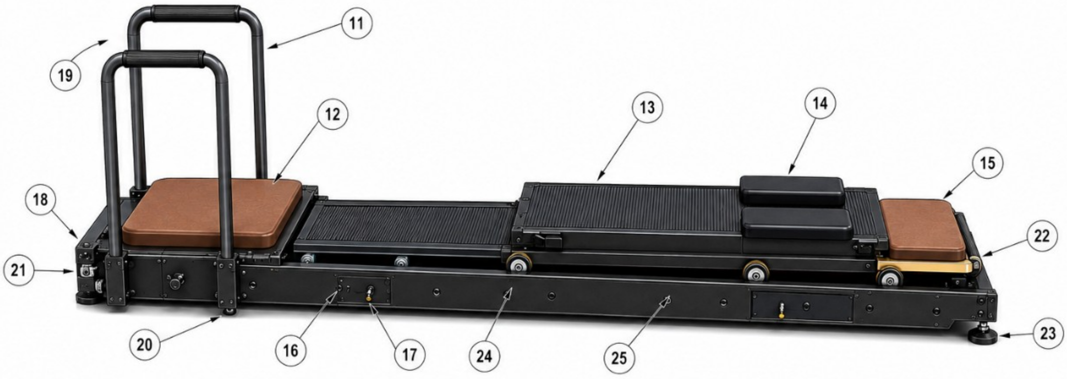
REFERENCE NUMERALS	
11	Vertical Support Rail (Left & Right)
12	Fixed Kneeling Pad
13	Sliding Platform Assembly
14	Forearm Support Pads
15	Hand Contact Surface
16	*Low-Friction Rolling Elements (Inside Channel)*
17	Longitudinal Channel Rail
18	Quick-Release Locking Knob
19	Rail Grip (Polymer)
20	Base Frame Chassis
21	Rear Travel Stop
22	Front Rubber Bumper
23	Adjustable Leveling Feet
24	Kinetic Restoration Subsystem (Inside)
25	Side Access Panel (Both Sides)

SPECIFICATIONS (Typical)	
Overall Length	~1620 mm
Overall Width	~600 mm
Overall Height (Rails)	~720 mm
Sliding Stroke	~1100 mm
Device Weight	~28–35 kg (varies by configuration)
Max User Load	~150 kg (safe working load)

Note: Visual proportions are indicative.
Actual dimensions and materials may vary.

FIG. 2 – SIDE PERSPECTIVE VIEW (LEFT SIDE)

Primary Components and Linear Motion Path

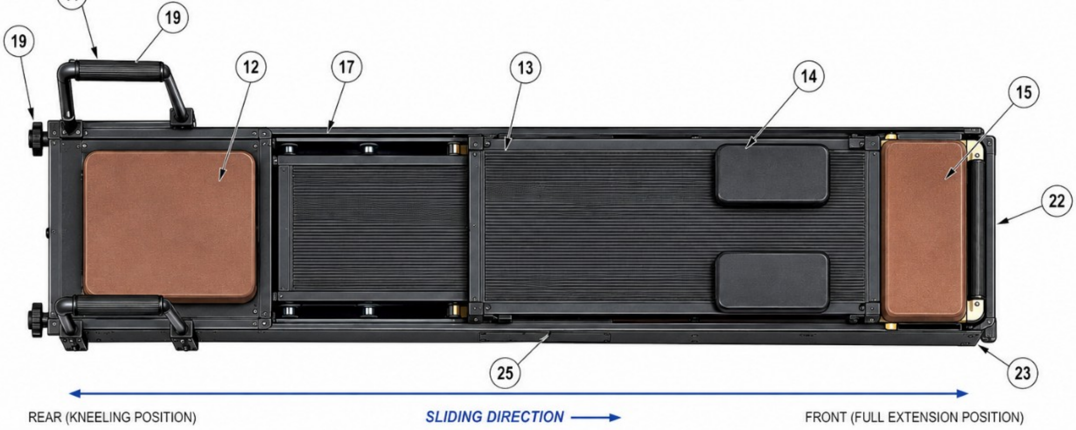


REAR (KNEELING POSITION) — *SLIDING STROKE ~ 1100 mm* — FRONT (FULL EXTENSION POSITION)

LINEAR MOTION PATH
The sliding platform assembly (13) translates linearly along the longitudinal channel rail (17) on low-friction rolling elements (16) from the rear kneeling position to the front full-extension position and returns under assisted force from the Kinetic Restoration Subsystem (24).

REFERENCE NUMERALS			
11	Vertical Support Rail (Left & Right)	19	Rail Grip (Polymer)
12	Fixed Kneeling Pad	20	Base Frame Chassis
13	Sliding Platform Assembly	21	Rear Travel Stop
14	Forearm Support Pads	22	Front Rubber Bumper
15	Hand Contact Surface	23	Adjustable Leveling Feet
16	*Low-Friction Rolling Elements (Inside Channel)*	24	Kinetic Restoration Subsystem (Inside)
17	Longitudinal Channel Rail	25	Side Access Panel (Both Sides)
18	Quick-Release Locking Knob		

FIG. 3 – TOP PLAN VIEW (LABELLED)



REFERENCE NUMERALS			
11	Vertical Support Rail (Left & Right)	18	Quick-Release Locking Knob
12	Fixed Kneeling Pad	19	Rail Grip (Polymer)
13	Sliding Platform Assembly	22	Front Rubber Bumper
14	Forearm Support Pads	23	Adjustable Leveling Feet
15	Hand Contact Surface	25	Side Access Panel (Both Sides)
16	*Low-Friction Rolling Elements (Inside Channel)*	20	Base Frame Chassis
		21	Rear Travel Stop
17	Longitudinal Channel Rail		

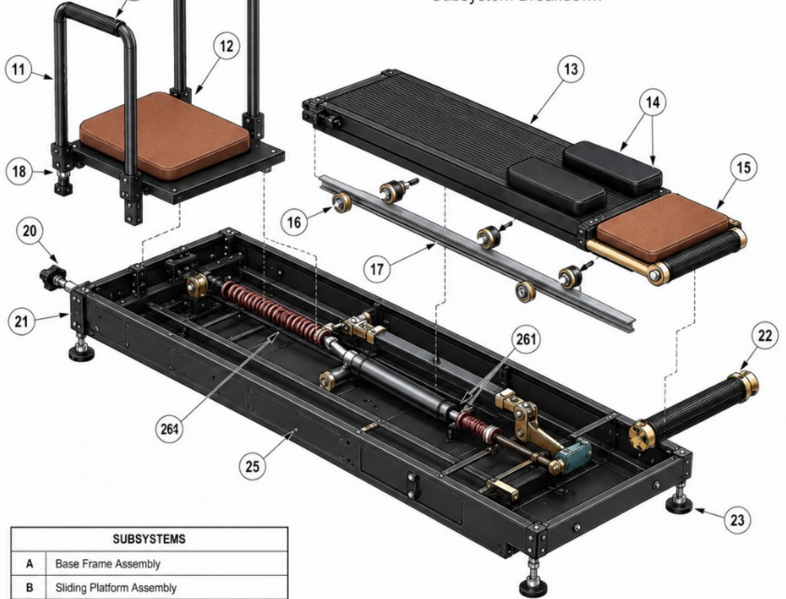
SPECIFICATIONS (Typical)	
Overall Length	~1620 mm
Overall Width	~600 mm
Sliding Stroke	~1100 mm
Overall Height (Rails)	~720 mm
Device Weight	~28–35 kg (varies by configuration)
Max User Load	~150 kg (safe working load)

NOTES:

- *All surfaces are shown in plan.*
- *Padding (pads) are shown in brown.*
- *Rail grips (19) shown with ribbed texture for clarity.*
- *Visual proportions are indicative.*
- *Actual dimensions and materials may vary.*

FIG. 4 – EXPLODED PERSPECTIVE VIEW

Subsystem Breakdown



REFERENCE NUMERALS	
11	Vertical Support Rail (Left & Right)
12	Fixed Kneeling Pad
13	Sliding Platform Assembly
14	Forearm Support Pads (Pair)
15	Hand Contact Surface
16	Low-Friction Rolling Elements (Inside Channel)
17	Longitudinal Channel Rail
18	Quick-Release Locking Knob
19	Rail Grip (Polymer)
20	Base Frame Chassis
21	Rear Travel Stop
22	Front Rubber Bumper
23	Adjustable Leveling Feet (4 Nos.)
25	Side Access Panel (Both Sides)
261	Extension Spring / Elastomeric Cord
264	Gas Compression Strut Cylinder
268	Cantilever Arm
269	Secondary Leverage Spring Component
267	Eccentric Cam Plate
266	Oblong Mounting Pivot Matrix
262	Adjustable Anchor Pin

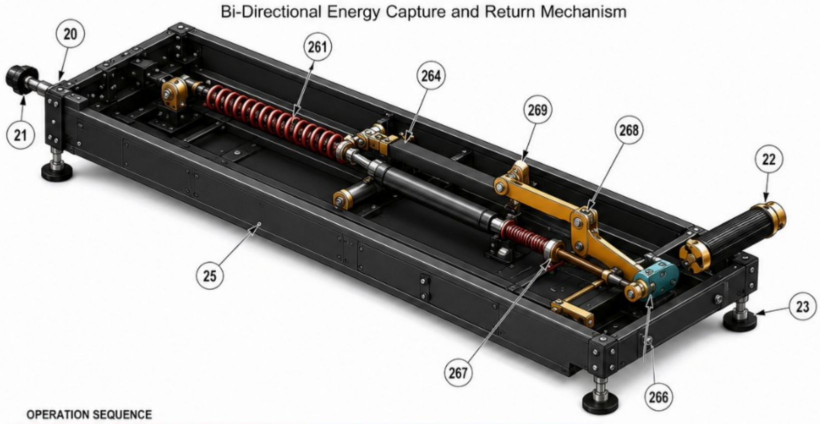
SUBSYSTEMS	
A	Base Frame Assembly
B	Sliding Platform Assembly
C	Bi-Directional Kinetic Restoration Subsystem
D	Support Rail Assembly
E	Electronic Monitoring Subsystem (Optional)

Note: All fasteners, screws and minor hardware are not shown for clarity.

KINETIC RESTORATION SUBSYSTEM (C) **Components Shown Inside Base Frame**	
261	Extension Spring / Elastomeric Cord
264	Gas Compression Strut Cylinder
269	Secondary Leverage Spring Component
267	Eccentric Cam Plate
268	Cantilever Arm

FIG. 5 – INSIDE VIEW OF KINETIC RESTORATION SUBSYSTEM (C)

Bi-Directional Energy Capture and Return Mechanism



REFERENCE NUMERALS – SUBSYSTEM (C)	
20	Base Frame Chassis (Part of Subsystem A)
21	Rear Travel Stop
22	Front Rubber Bumper
23	Adjustable Leveling Foot
25	Side Access Panel (Removed for Clarity)
261	Extension Spring / Elastomeric Cord
264	Gas Compression Strut Cylinder
268	Cantilever Arm
269	Secondary Leverage Spring Component
267	Eccentric Cam Plate
266	Oblong Mounting Pivot Matrix
262	Adjustable Anchor Pin

→ Forward Stroke (Energy Capture)
Spring (261) + Strut (264) are compressed

→ **Return Stroke (Energy Release)**
Stored energy assists rearward motion

OPERATION SEQUENCE

A. FORWARD STROKE – ENERGY CAPTURE

User moves forward

- As the sliding platform moves forward, the cantilever arm (268) rotates about the oblong pivot matrix (266).
- The extension spring / elastomeric cord (261) is stretched.
- The gas compression strut (264) is compressed.
- Kinetic energy of the user is stored in both elements.

B. RETURN STROKE – ENERGY RELEASE

Assisted return to kneeling position

- Upon release of forward force, the stored energy in spring (261) and strut (264) generates rearward assistance.
- The cantilever arm (268) and leverage component (269) convert and transmit force to pull the sliding platform rearward.
- Resistance profile can be tuned via cam plate (267) and anchor pin (262) positions.

ECCENTRIC CAM PLATE (267) – RESISTANCE TUNING

Moving the anchor pin (262) to different holes on the cam plate changes the effective leverage ratio and resistance curve.

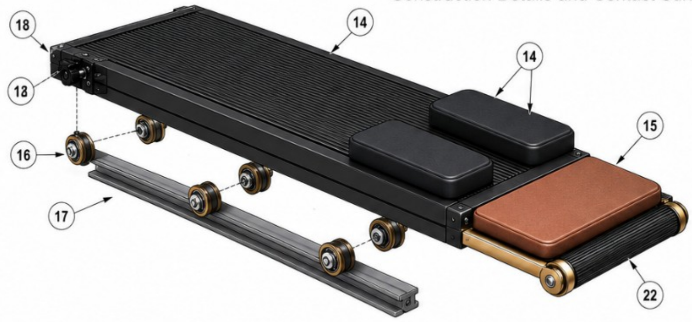
EFFECT OF PIN POSITION

- **Hole A** – Higher Assistance
  Lower resistance during forward stroke
- **Hole B** – Medium Assistance
  Balanced resistance
- **Hole C** – Lower Assistance
  Higher resistance during forward stroke

NOTE: The Kinetic Restoration Subsystem (C) operates passively and requires no electrical power.
All components are maintenance-free or low-maintenance for long service life.

FIG. 6 – SLIDING PLATFORM ASSEMBLY (SUBSYSTEM B)

Construction Details and Contact Surfaces

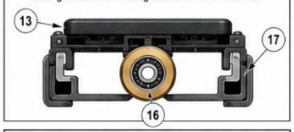


REFERENCE NUMERALS – SUBSYSTEM B	
13	Sliding Platform Assembly (Main Deck)
14	Forearm Support Pads (Pair)
15	Hand Contact Surface
16	Low-Friction Rolling Elements (Inside Channel)
17	Longitudinal Channel Rail (Mounted in Base)
18	Quick-Release Locking Knob
22	Front Rubber Bumper

TOP VIEW – CONTACT SURFACES

CROSS-SECTION VIEW (SECTION A–A)

Sliding Platform on Rolling Elements within Channel



ROLLING ELEMENT SPECIFICATIONS (Typical)	
Type	Sealed Ball Bearing Wheel
Wheel Diameter	~40 mm
Bearing Type	2RS Sealed
Material	PU (Polyurethane) Tread
Load Capacity (Each)	~80 kg (Dynamic)

CONTACT SURFACE MATERIALS

Forearm Support Pads (14)	Hand Contact Surface (15)	Main Deck Surface (15)
		
• Material: Closed-Cell EVA Foam • Density: ~60 kg/m³ • Surface: Fine Texture, Non-Slip • Function: Distributes forearm load and reduces pressure on elbows	• Material: High-Density EVA Foam • Density: ~80 kg/m³ • Surface: Anti-Slip, Skin-Friendly • Function: Cushions palms and reduces impact during extension	• Material: Glass-Filled Nylon / ABS • Surface: Ribbed Anti-Slip Texture • Function: Provides stable support and low-friction environment

NOTE:
All contact surfaces are replaceable modules secured with concealed fasteners for maintenance and hygiene.

QUICK-RELEASE LOCKING KNOB (18) & TRAVEL FEATURES

Pull out knob (18), slide platform to desired position, release to lock.

Sliding travel is limited at the rear by the travel stop (21) and at the front by the rubber bumper (22) to prevent over-extension and provide a soft impact.

FUNCTION: The sliding platform assembly (13) provides a smooth, low-friction forward path for the upper body, carrying forearm and hand load while minimizing impact and shear on joints.

FIG. 7 – SUPPORT RAIL ASSEMBLY (SUBSYSTEM D)

Ergonomic Grip, Stability and Alignment

REFERENCE NUMERALS – SUBSYSTEM D	
11	Vertical Support Rail (Left & Right)
12	Fixed Kneeling Pad (Part of Subsystem A)
18	Quick-Release Locking Knob
19	Rail Grip (Polymer)
20	Base Frame Chassis
23	Adjustable Leveling Foot
30	Vertical Support Rail Base / Bracket
31	Crossbar (Rail Top Connector)
32	Pin-Locking Adjustment Matrix

FRONT VIEW (REAR END OF DEVICE)

RAIL HEIGHT ADJUSTMENT (Pin-Locking)

LOW POSITION (Elderly / Rehab)

HIGH POSITION (Standard / Tall Users)

FEATURES & FUNCTIONS

- Provides stable, vertical handholds during descent and return phases.
- Reduces load on wrists and shoulders by enabling bodyweight transfer through the arms.
- Height adjustable to accommodate users of different stature and mobility levels.
- Crossbar (31) adds torsional rigidity and provides optional reference alignment.
- Quick-release knobs (18) allow rapid folding or removal for transport.

MATERIAL SPECIFICATIONS (Typical)

Rail Tubes (11)	Aluminum Alloy 6061-T6
Rail Grip (19)	Thermoplastic Rubber (TPE)
Base Brackets (30)	Mild Steel / Powder Coated
Crossbar (31)	Aluminum Alloy
Locking Knob (18)	Glass-Filled Nylon
Pin-Locking Matrix (32)	Stainless Steel Insert

NOTES: Rails can be independently removed or folded by loosening knobs (18) and lifting upward.
All edges are radiused; surfaces are powder-coated or anodized for durability and silent operation.

FIG. 8 – OPERATION SEQUENCE OF APD

Full Prostration Cycle with Kinetic Restoration Assistance

STAGE 1 – KNEELING START POSITION

- User kneels on fixed pad (12).
- Hands rest on hand contact surface (15).
- Body upright, system at rear (home) position.

SYSTEM STATE

- Springs/strut (261, 264) in neutral.
- No stored energy.
- Platform at rear stop (21).

STAGE 2 – FORWARD EXTENSION (ENERGY CAPTURE)

- User glides forward to full extension.
- Body weight drives platform forward.
- Springs (261) and/or strut (264) are compressed.
- Kinetic energy is stored.

SYSTEM STATE

- Forward stroke.
- Energy captured and stored in KRS (C).
- Maximum compression near full extension.

STAGE 3 – TRANSITION (DIRECTION CHANGE)

- At full extension, user pauses briefly (doctrinal position).
- Begins to pull body backward to return.
- Platform reverses direction.

SYSTEM STATE

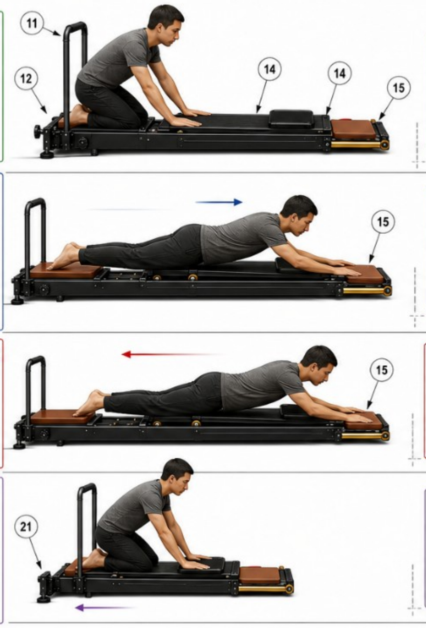
- Direction reverses.
- Stored energy begins to assist return.
- Load transfers from forward pressure to rearward pull.

STAGE 4 – RETURN (ENERGY RELEASE)

- Stored energy in springs (261) and/or strut (264) assists pulling platform rearward.
- User returns to kneeling position with reduced effort.
- Platform reaches rear stop (21).

SYSTEM STATE

- Energy released to assist return.
- System returns to neutral state.
- Cycle complete.



ONE FULL PROSTRATION CYCLE COMPLETE

Repeat sequence for continuous practice.

REFERENCE NUMERALS – KEY COMPONENTS	
11	Vertical Support Rail (Left & Right)
12	Fixed Kneeling Pad
13	Sliding Platform Assembly (Main Deck)
14	Forearm Support Pads (Pair)
15	Hand Contact Surface
21	Rear Travel Stop
261	Extension Spring / Elastomeric Cord
264	Gas Compression Strut Cylinder

LEGEND

- → Stage 1 – Start / Neutral
- → Stage 2 – Forward Stroke (Energy Capture)
- → Stage 3 – Transition / Direction Change
- → Stage 4 – Return Stroke (Energy Release)
- ¦ Travel Limits (Rear: 21 / Front: 22)

Kinetic Energy Flow

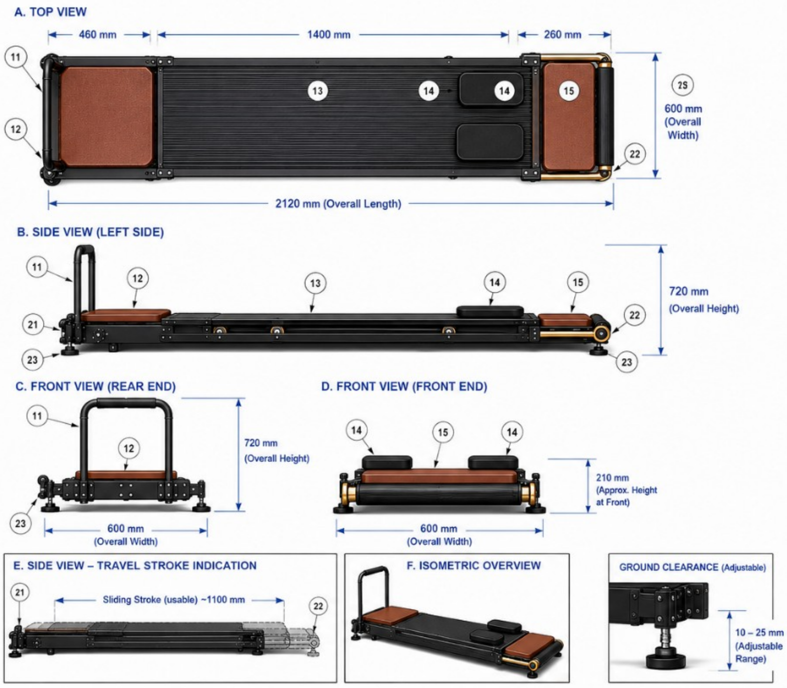
- ○ Neutral (No Energy)
- → Energy Capture (Store)
- - - → Transition
- → Energy Release (Return Assist)

NOTES

- Assistance level is configurable via resistance tuning (see Fig. 5 – Cam Plate 267 & Anchor Pin 262).
- The APD preserves the complete doctrinal movement pattern without modification.
- Operation is silent, passive, and requires no external electrical power.
- User still performs the effort; the device reduces cumulative joint loading and provides ascent assistance.

FIG. 9 – OVERALL DIMENSIONS AND ORTHOGRAPHIC VIEWS

Assisted Prostration Device (APD) – Typical Configuration



NOTES

- Dimensions are typical and may vary based on user requirements.
- All dimensions in millimeters (mm).
- Tolerances: ±5 mm unless stated otherwise.

REFERENCE NUMERALS – KEY COMPONENTS

No.	Component
11	Vertical Support Rail (Left & Right)
12	Fixed Kneeling Pad
13	Sliding Platform Assembly (Main Deck)
14	Forearm Support Pads (Pair)
15	Hand Contact Surface
21	Rear Travel Stop
22	Front Rubber Bumper
23	Adjustable Leveling Feet
20	Base Frame Chassis

DIMENSION SUMMARY (Typical)

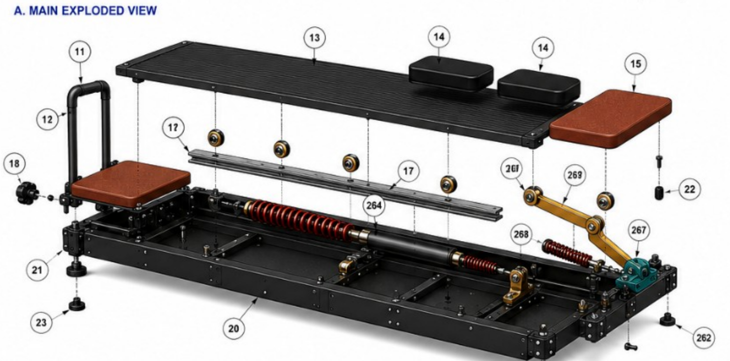
Parameter	Dimension
Overall Length	2120 mm
Overall Width	600 mm
Overall Height (to top of rail)	720 mm
Sliding Stroke Length (usable)	~1100 mm
Kneeling Pad Size (L × W × T)	460 × 600 × 40 mm
Forearm Pad Size (each)	250 × 120 × 40 mm
Hand Pad Size	260 × 600 × 40 mm
Platform Height from Floor	~85–95 mm (adjustable)

MATERIALS (Typical)

Component	Material / Finish
Base Frame Chassis (20)	Powder Coated Steel
Sliding Deck (13)	Anodized Aluminum
Rails (11)	Aluminum Alloy / Powder Coated
Pads (12, 14, 15)	EVA Foam / PU Cover
Bumper (22)	Natural Rubber
Leveling Feet (23)	Steel & Rubber
Fasteners	Stainless Steel

FIG. 10 – EXPLODED VIEW AND COMPONENT BREAKDOWN

Assisted Prostration Device (APD)



REFERENCE NUMERALS – ALL COMPONENTS

No.	Component	No.	Component
11	Vertical Support Rail (Left & Right)	261	Extension Spring / Elastomeric Cord
12	Fixed Kneeling Pad	264	Gas Compression Strut Cylinder
13	Sliding Platform Assembly (Main Deck)	267	Eccentric Cam Plate
14	Forearm Support Pads (Pair)	268	Cantilever Arm
15	Hand Contact Surface	269	Secondary Leverage Spring Component
16	Low-Friction Rolling Elements	262	Adjustable Anchor Pin
17	Longitudinal Channel Rail	266	Oblong Mounting Pivot Matrix (Not visible)
18	Quick-Release Locking Knob	30	Vertical Support Rail Base Bracket (Not shown)
20	Base Frame Chassis (Subsystem A)	31	Crossbar (Rail Top Connector) (Not shown)
21	Rear Travel Stop	32	Pin-Locking Adjustment Matrix (Not shown)
22	Front Rubber Bumper	40	Electronic Monitoring Module (Optional) (Not shown)
23	Adjustable Leveling Foot		
25	Side Access Panel (Not shown)		(Additional fasteners listed below)

B. SUB-ASSEMBLY GROUPS

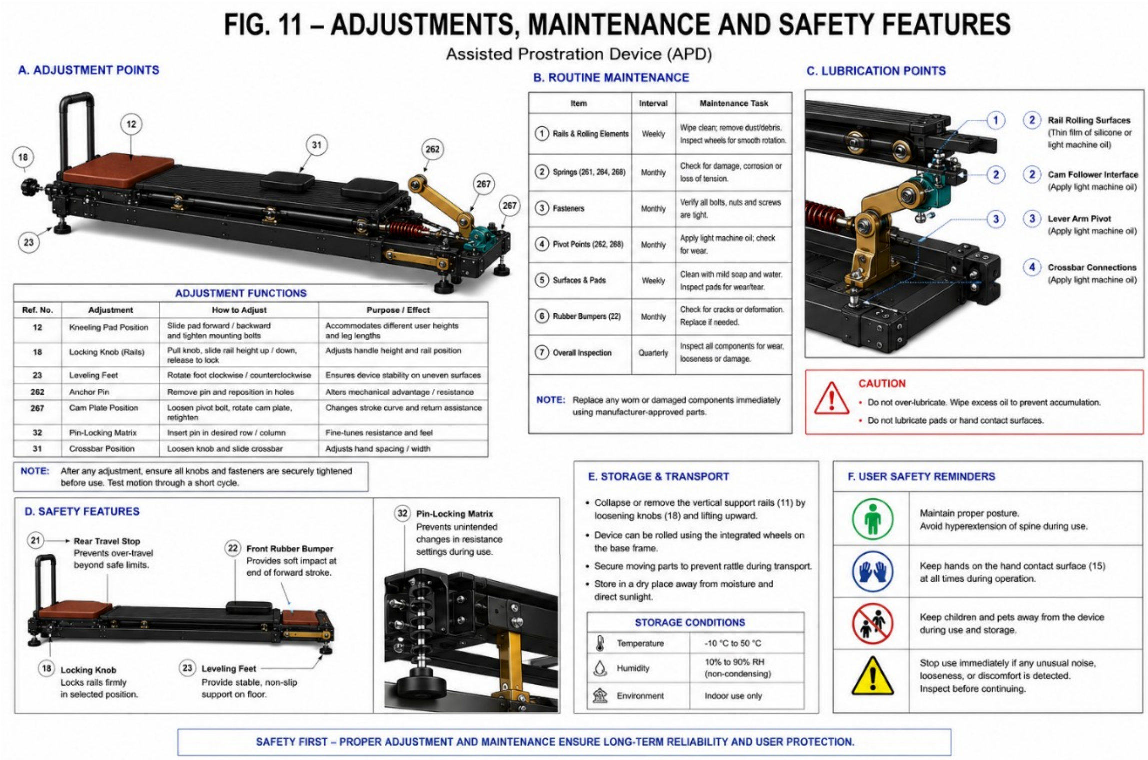
C. FASTENERS & SMALL PARTS (TYPICAL)

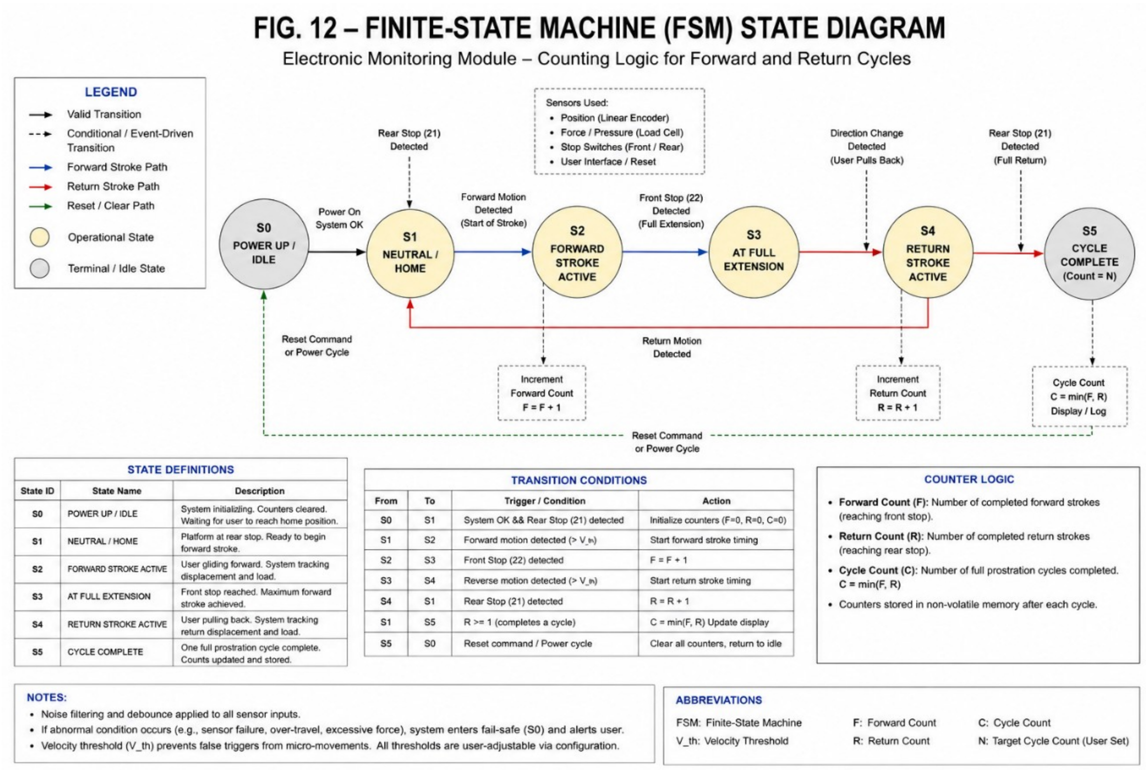
D. MATERIAL SUMMARY (TYPICAL)

Component Group	Material / Finish
Base Frame Chassis	Powder Coated Steel
Sliding Deck & Rails	Anodized Aluminum Alloy
Support Rails & Crossbar	Aluminum Alloy / Powder Coated
Pads (Kneeling / Forearm / Hand)	EVA Foam / PU Cover
Rolling Elements (Wheels)	PU (Polyurethane) on Sealed Ball Bearings
Springs	High Carbon Steel
Gas Strut Cylinder	Steel / Painted
Cam Plate	Aluminum Alloy
Fasteners	Stainless Steel
Bumpers	Natural Rubber
Knobs	Nylon / Glass-Filled Nylon

NOTES:

- Exploded view is for illustrative purposes to show relative positions and assembly order.
- All components are designed for easy disassembly and maintenance.
- Fasteners are predominantly metric standard.
- Optional modules can be omitted without affecting core mechanical function.





12. APPENDIX — FIGURE 11: FSM STATE DIAGRAM (TEXT REPRESENTATION)

Finite-state machine logic for the Electronic Monitoring Module (40) as described in Section 6.6 and claimed in Claim 7.

```
+----------------------------------------------------+
```

```
|                   STATE 0                         |
|       Baseline / Kneeling / Standing Stance       |
|   FSR Array (42) Pressure: HIGH  ( > P_high )     |
+---------------------------------------------------+
                        |
    Practitioner extends forward to prone position;
    Kneeling pad (12) pressure drops below P_low
                        |
                        v
+---------------------------------------------------+
|                   STATE 1                         |
|         Prone / Forward Extension Phase           |
|   FSR Array (42) Pressure: LOW   ( < P_low )      |
|      [ Forward Extension Phase Validated ]        |
+---------------------------------------------------+
                        |
    Practitioner initiates ascent return;
    Kneeling pad (12) pressure recovers above P_high
                        |
                        v
+---------------------------------------------------+
|                   STATE 2                         |
|        Ascent Recovery / Return to Kneeling       |
|   FSR Array (42) Pressure: HIGH  ( > P_high )     |
|       [ Ascent Return Phase Validated ]           |
+---------------------------------------------------+
                        |
    State sequence 0 -> 1 -> 2 confirmed.
    MCU (44) executes count increment.
    Display (46) updated.
                        |
                        v
           [ CYCLE COUNT = COUNT + 1 ]
                        |
                        +-----> Returns to STATE 0
```

Non-sequential pressure events — including knee repositioning during rest, robe or clothing adjustment, partial extension, or accidental contact — fail to satisfy the consecutive 0-to-1-to-2 state sequence and are discarded as noise by the MCU (44). This logic ensures valid count registration exclusively for complete, full-range prostration cycles.

13. STATEMENT OF NOVELTY

The novelty of the present invention resides in the combination of: a constrained linear sliding platform assembly (13) with approximately 1100 mm usable stroke, optimised for the floor-level prone-extension geometry specific to full-body devotional prostration cycles; a bi-directional kinetic restoration subsystem (24) providing partial passive return assistance calibrated to the eccentric loading profile of high-repetition kneeling-to-prone-to-kneeling sequences across multiple embodiments (C1 through C5); configurable resistance geometry via eccentric cam plate (267) with selectable hole positions enabling user-specific tuning; safety features including travel stops (21, 22), adjustable levelling feet (23), and damped return speed; and an optional finite-state machine electronic counting module (40) providing reliable cycle enumeration immune to false triggers — all in an integrated non-wearable floor-based apparatus of approximately 2120 mm (L) x 600 mm

(W) x 720 mm (H) that preserves the complete doctrinal motion sequence without abbreviation. No prior art discloses this combination for this use case.

DECLARATION

I, Joy Bose, hereby declare that I am the original designer of the Assisted Prostration Device described in this specification, and that I am releasing this design as open hardware under the CERN Open Hardware Licence Version 2 — Strongly Reciprocal (CERN OHL-S v2) for the free use of all practitioners, makers, and engineers globally.

Signature: ______________________________

Name (Print): Joy Bose

Date: May 2026

Place: Bengaluru, Karnataka, India

--- END OF OPEN HARDWARE SPECIFICATION v1.0 ---